F70LA Life Insurance Mathematics A (2025/26)
Tutorial 2 Solutions

1. The probability that:

   (i) A life currently aged 30 will die before exact age 60 is ${}_{30}q_{30}$.

   (ii) A life currently aged 50 will survive to at least age 80 is ${}_{30}p_{50}$.

   (iii) A life currently aged 40 will die between exact ages 55 and 65 is ${}_{15}|_{10}q_{40}$.

   (iv) A life currently aged 30 will die either between exact ages 40 and 45 or between exact ages 50 and 55 is ${}_{10}|_{5}q_{30} + {}_{20}|_{5}q_{30}$.

2. (i) ${}_{40}p_0 = \mathbb{P}[T_0 > 40] = S_0(40) = 1 - F_0(40) = 0.88976$.

   (ii) ${}_{20}p_{60} = \mathbb{P}[T_{60} > 20] = S_{60}(20) = 1 - F_{60}(20) = 0.90087$.

   (iii) ${}_{5}|_{5}q_{50} = \mathbb{P}[5 \leq T_{50} < 10] = F_{50}(10) - F_{50}(5) = 0.02246$.

3. With $p_x = \frac{89-x}{10}$,

   (i) $q_{80} = 1 - p_{80} = 1 - 0.9 = 0.1$.

   (ii) ${}_2p_{80} = p_{80}\, p_{81} = (0.9)(0.8) = 0.72$.

   (iii) ${}_5p_{80} = \prod_{k=80}^{84} p_k = (0.9)(0.8)(0.7)(0.6)(0.5) = 0.1512$.

   (iv) ${}_3q_{82} = 1 - {}_3p_{82} = 1 - p_{82}\, p_{83}\, p_{84} = 1 - (0.7)(0.6)(0.5) = 0.79$.

   (v) ${}_2|_3q_{80} = {}_2p_{80}\, {}_3q_{82} = (0.72)(0.79) = 0.5688$.

4. Since the force of morality has a different form at age 40, the integral has to be split at the time corresponding to age 40.

   (i)

$$
\begin{aligned}
S_0(50) = \exp\left(-\int_0^{50} \mu_s ds\right) &= \exp\left(-\int_0^{40} \mu_s ds - \int_{40}^{50} \mu_s ds\right) \\
&= \exp\left(-\int_0^{40} \mu_s ds\right) \times \exp\left(-\int_{40}^{50} \mu_s ds\right) \\
&= \exp\left(-\int_0^{40} 0.001 ds\right) \times \exp\left(-\int_{40}^{50} 0.001 + 0.01(s-40) ds\right) \\
&= \exp\left(-0.04\right) \times \exp\left(3.99 - 0.01\left[0.5s^2\right]_{40}^{50}\right) \\
&= \exp\left(-0.04\right) \times \exp\left(3.99 - 4.5\right) = \exp(-0.55) = 0.57695.
\end{aligned}
$$

(ii)

$$\begin{aligned}
{}_{12}q_{35} =& 1 - {}_{12}p_{35} = 1 - {}_{5}p_{35} \times {}_{7}p_{40} \\
=& 1 - \exp\left(-\int_{35}^{40} \mu_s ds\right) \times \exp\left(-\int_{40}^{47} \mu_s ds\right) \\
=& 1 - \exp\left(-\int_{35}^{40} 0.001 ds\right) \times \exp\left(-\int_{40}^{47} 0.001 + 0.01(s-40) ds\right) \\
=& 1 - \exp\left(-0.005\right) \times \exp\left(0.399 - \left[0.005 s^2\right]_{40}^{47}\right) \\
=& 1 - \exp\left(-0.005\right) \times \exp\left(-0.252\right) = 1 - 0.773368 = 0.22663
\end{aligned}$$

5. Either, using the fundamental theorem of calculus:

$$\begin{aligned}
\frac{\partial}{\partial x} {}_{t}p_x &= \frac{\partial}{\partial x}\left[\exp\left(-\int_{x}^{x+t} \mu_s ds\right)\right] = \exp\left(-\int_{x}^{x+t} \mu_s ds\right) \cdot \frac{\partial}{\partial x}\left(-\int_{x}^{x+t} \mu_s ds\right) \\
&= \exp\left(-\int_{x}^{x+t} \mu_s ds\right) \cdot \frac{\partial}{\partial x}\left(-\left(\int_{0}^{x+t} \mu_s ds + \int_{x}^{0} \mu_s ds\right)\right) \\
&= \exp\left(-\int_{x}^{x+t} \mu_s ds\right) \cdot \frac{\partial}{\partial x}\left(-\int_{0}^{x+t} \mu_s ds + \int_{0}^{x} \mu_s ds\right) \\
&= {}_{t}p_x \left(\mu_x - \mu_{x+t}\right),
\end{aligned}$$

or, using the quotient rule of differentiation:

$$\begin{aligned}
\frac{\partial}{\partial x} {}_{t}p_x = \frac{\partial}{\partial x} S_x(t) &= \frac{\partial}{\partial x}\left(\frac{S_0(x+t)}{S_0(x)}\right) \\
&= \frac{S_0(x)\frac{\partial}{\partial x} S_0(x+t) - (\frac{\partial}{\partial x} S_0(x)) S_0(x+t)}{S_0(x)^2} \\
&= \frac{S_0(x)(-\mu_{x+t} S_0(x+t)) - (-\mu_x S_0(x)) S_0(x+t)}{S_0(x)^2} \\
&= S_x(t)(\mu_x - \mu_{x+t}) \\
&= {}_{t}p_x(\mu_x - \mu_{x+t}).
\end{aligned}$$

6. (i)

$${}_{10}p_{50} = \exp\left(-\int_{50}^{60} \mu_s ds\right) = \exp\left(-\int_{50}^{60} 0.03 ds\right) = e^{-0.3} = 0.74082.$$

(ii)

$$\begin{aligned} {}_{10}p_{70} &= \exp\left(-\int_{70}^{80}\mu_s ds\right) = \exp\left(-\int_{70}^{80}[0.03+0.003(s-70)]ds\right) \\ &= \exp\left(\int_{70}^{80}[0.18-0.003s]ds\right) = \exp\left(\left[0.18s-0.0015s^2\right]_{70}^{80}\right) \\ &= \exp(-0.45) = 0.63763. \end{aligned}$$

(iii)

$$\begin{aligned} {}_{7}p_{70} &= \exp\left(-\int_{70}^{77}\mu_s ds\right) = \exp\left(-\int_{70}^{77}\left(0.03+0.003(s-70)\right])\,ds\right) \\ &= \exp\left(0.18s-\left[0.0015s^2\right]_{70}^{77}\right) = \exp(-0.2835) = 0.75314. \end{aligned}$$

Therefore ${}_{7}q_{70} = 1 - {}_{7}p_{70} = 0.24686$.

(iv) First, ${}_{8}|{}_{7}q_{66} = {}_{8}p_{66} - {}_{15}p_{66} = {}_{4}p_{66}\cdot{}_{4}p_{70} - {}_{4}p_{66}\cdot{}_{11}p_{70}$.
Then calculate ${}_{4}p_{66} = e^{-4(0.03)} = 0.88692$,

$${}_{4}p_{70} = \exp\left(\left[0.18s-0.0015s^2\right]_{70}^{74}\right) = e^{-0.1440} = 0.86589$$

and

$${}_{11}p_{70} = \exp\left(\left[0.18s-0.0015s^2\right]_{70}^{81}\right) = e^{-0.5115} = 0.59960.$$

Therefore, ${}_{8}|{}_{7}q_{66} = 0.88692(0.86589-0.59960) = 0.23618$.

7. If ${}_{t}p_x\mu_{x+t}$ is strictly increasing on $0 \le t \le 1$, then for any $0 < s \le 1$, ${}_{s}p_x\mu_{x+s} > {}_{0}p_x\mu_{x+0} = \mu_x$, as ${}_{0}p_x = 1$.

Recall that ${}_{t}q_x = \int_0^t {}_{r}p_x\mu_{x+r}\,dr$. Therefore, with $t=1$, $q_x = \int_0^1 {}_{r}p_x\mu_{x+r}\,dr$. Applying the inequality, ${}_{s}p_x\mu_{x+s} > mu_x$,

$$q_x = \int_0^1 {}_{r}p_x\mu_{x+r}dr > \int_0^1 \mu_x dr = \mu_x\int_0^1 dr = \mu_x.$$

8. (i) In this model, for $t \ge 0$, $S_x(t) = {}_{t}p_x = e^{-\lambda t}$. Since

$$\lim_{t\to\omega}\{S_x(t)\} = \lim_{t\to\omega}\left\{e^{-\lambda t}\right\} = 0,$$

is satisfied only by $\omega = \infty$ then, by the conditions required to be satisfied by a survival function, the limiting age is ∞.

(ii)
$$\mathring{e}_x = \int_0^\infty {}_tp_x dt = \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}.$$

(iii)
$$e_x = \sum_{k=1}^\infty {}_kp_x = \sum_{k=1}^\infty e^{-\lambda k} = \frac{e^{-\lambda}}{1-e^{-\lambda}}.$$

(iv) From part (ii), choosing $\lambda = 0.1$ results in $\mathring{e}_{75} = 10$.

9. As all q_x are identical, so are the p_x. Hence, ${}_tp_x = p_x p_{x+1} \cdots p_{x+t-1} = (p_x)^t$.

(i) ${}_{55}p_{30} = (p_{30})^{55} = (1-0.02)^{55} = 0.3292$.

(ii) ${}_{25}p_{50} = (p_{50})^{25} = 0.98^{25} = 0.6035$.

(iii) ${}_{15}q_{30} = 1 - {}_{15}p_{30} = 1 - (0.98)^{15} = 0.2614$.

(iv) ${}_{10}|_3q_{40} = {}_{10}p_{40} - {}_{13}p_{40} = (0.98)^{10} - (0.98)^{13} = 0.0481$.

10.
$$S_x(t) = {}_tp_x = \exp\left(-\int_x^{x+t} \mu_s ds\right) = \exp\left(-\int_x^{x+t} 0.02s\, ds\right) = e^{-0.01t(2x+t)}.$$

Therefore the p.d.f. of T_x is
$$-\frac{d}{dt}S_x(t) = -\frac{d}{dt}(e^{-0.01t(2x+t)}) = 0.02(x+t)e^{-0.01t(2x+t)}.$$